

Case Studies

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[Practical]

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_task4
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_task4
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ touch public.txt private.txt script.s
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano public.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano private.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ nano script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod 600 private.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod 644 public.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ chmod +x script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ./script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ ls -l
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 10 15:32 private.txt
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 10 15:32 public.txt
-rwxr-xr-x 1 angela_jose angela_jose 0 Aug 10 15:32 script.sh
angela_jose@LAPTOP-B3CJ092Q:~/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories

```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir permission_lab
angela_jose@LAPTOP-B3CJ092Q:~$ cd permission_lab
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ touch test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ nano test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u-r test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
--w-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u+r test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-rw-r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u-w test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ nano test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ chmod u+w text.txt
chmod: cannot access 'text.txt': No such file or directory
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ls -l
total 4
-r--r--r-- 1 angela_jose angela_jose 17 Aug 10 15:38 test.txt
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
angela_jose@LAPTOP-B3CJ092Q:~/permission_lab$ |

```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp

```

angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_task6
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_task6
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ touch important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ nano important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mkdir backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp important.txt backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important.txt
└── important.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ rm important.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 1 file
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt
cp: missing destination file operand after 'backup/important_backup.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt .
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup  important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

```

```

├── important_backup.txt

2 directories, 1 file
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt
cp: missing destination file operand after 'backup/important_backup.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cp backup/important_backup.txt .
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ ls
backup  important_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ tree
.
├── backup
│   └── important_backup.txt
└── important_backup.txt

2 directories, 2 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ mv important_backup.txt important_restored.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ cat important_restored.txt
Welcome to Linux
angela_jose@LAPTOP-B3CJ092Q:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir linux_project
angela_jose@LAPTOP-B3CJ092Q:~$ cd linux_project
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ mkdir source backup docs
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ touch source/main.c source/helper.c source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls source/
README.txt helper.c main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ nano source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ nano source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ head -5 source/README.txt
Operating System Lab
Linux Commands
File Mangement
Directory Management
Permissions
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tail -5 source/README.txt
Docs Folder
Learning Commands
Operating Systems
End of File
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tree
.
├── backup
├── docs
└── source
    ├── README.txt
    ├── helper.c
    └── main.c

4 directories, 3 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/main.c backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup/
main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
backup docs source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt
cp: missing destination file operand after 'source/README.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt docs/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls docs
README.txt
```

```
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/main.c backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup/
main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
backup docs source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt
cp: missing destination file operand after 'source/README.txt'
Try 'cp --help' for more information.
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ cp source/README.txt docs/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls docs
README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ mv source/helper.c source/functions.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls source
README.txt functions.c main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ chmod 644 source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 angela_jose angela_jose 159 Aug 11 15:52 source/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ rm backup/main.c
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls backup
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ rmdir backup/
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ ls
docs source
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
angela_jose@LAPTOP-B3CJ092Q:~/linux_project$ |
```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise

```
angela_jose@LAPTOP-B3CJ092Q:~$ mkdir student_portal
angela_jose@LAPTOP-B3CJ092Q:~$ cd student_portal
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ mkdir src docs backup logs config
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls
backup  config  docs  logs  src
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tree
.
├── backup
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

6 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ nano logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ head -5 logs/system.log
head: cannot open 'logs/system.log' for reading: No such file or directory
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ head -5 logs/system.log
System Started
User Login
Configuration Loaded
Database Connected
Service Started
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tail -5 logs/system.log
Permissions Verified
File Synced
System Healthy
System Shutdown
Memory Checked
```

```
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ nano logs/system.log
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tail -5 logs/system.log
Permissions Verified
File Synced
System Healthy
System Shutdown
Memory Checked
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ cp docs/README.txt backup/
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ mv backup/README.txt backup/README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ chmod 600 config/config.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls -l config
total 0
-rw----- 1 angela_jose angela_jose 0 Aug 11 16:05 config.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ chmod 644 docs/README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls -l docs
total 0
-rw-r--r-- 1 angela_jose angela_jose 0 Aug 11 16:05 README.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ rm backup/README_backup.txt
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls backup
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ rmdir backup
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ ls
config  docs  logs  src
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
angela_jose@LAPTOP-B3CJ092Q:~/student_portal$ |
```